

3.3 Chamfered (knife-edge) boundaries

Every boundary that defines an edge, disk, or cutout must be chamfered from the back. Then no camera or projector ray can strike the boundary's side wall, and the sensor sees only the front face and the back plate. An unchamfered wall would be seen as a third surface by one camera and not the other, which corrupts exactly the edge measurements this procedure is meant to make.

- Cutouts and the square window: countersink from the back face, so the hole widens away from the sensor.
- Disks and the raised square: bevel the back, so the part narrows away from the sensor (a frustum).
- Land: the flat land left at the front edge must be at most 0.2 mm (EDGE_LAND_MAX_MM).
- Bevel angle: measured from the plate normal, it must exceed the worst-case ray angle plus a margin:

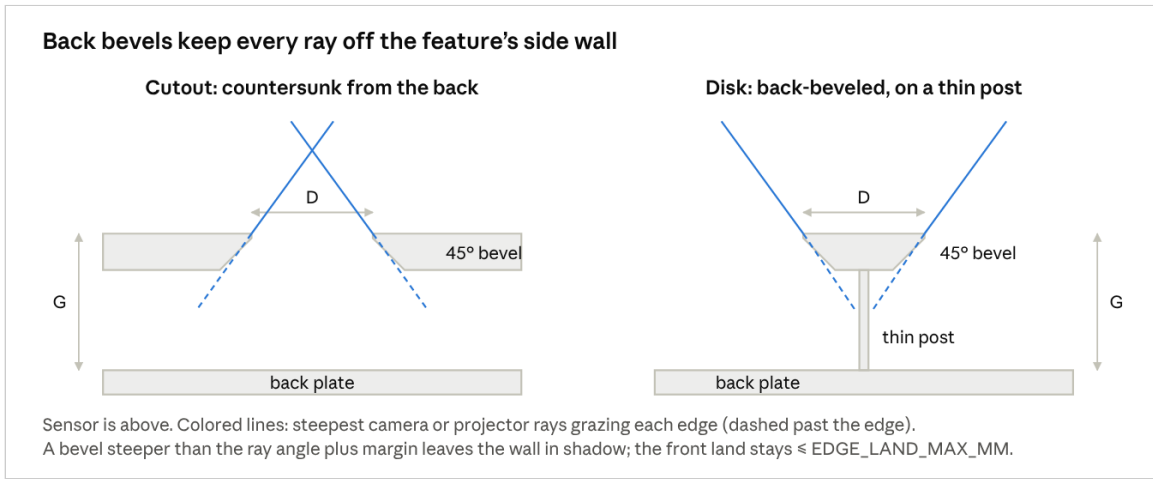


Figure 2. Chamfered cutout and disk, cross-section (not to scale)

In Figure 2 the steepest ray past each edge travels through open space and never meets the beveled wall, so the sensor sees only the front face and the back plate.

$$\theta_{\text{bevel}} \geq \max_{c \in \{L, R, P\}} \max_{e, \text{pose}} \angle(\mathbf{e} - \mathbf{c}, \hat{\mathbf{n}}_{\text{plate}}) + \theta_{\text{margin}}$$

Here c is the left camera, right camera, or projector center, e is any edge point at any pose used, and θ_{margin} is the 10° margin (CHAMFER_MARGIN_DEG). Chamfered targets are never tilted (the tilt sub-series uses T2), so the maximum runs over fronto-parallel poses; they are off-axis only in the C

field sub-series at the reference station, 800 mm (Z_REFERENCE_MM), and at the near limit they are centered. A worked value at the indicative geometry ($f_x \approx 688$ px, 640×480 px, 75 mm baseline, all unconfirmed): the raised square centered at the near limit gives about 24° , and the plates at the off-axis field positions at 800 mm about 29° ; with the 10° margin, 39° . A standard 45° bevel (90° included countersink) then passes; the 45° in Figure 2 is this example, not a requirement. Recompute this once the VSX3000 geometry is known. If the value exceeds 45° , use a steeper bevel or reduce the 0.6 field offset (FIELD_OFFSET_FRACTION) for the C, D, and E series.

Disk support posts. The post must sit behind the disk and be thinner than half (POST_DIAMETER_FRACTION_OF_D0) of the expected D_0 of Section 3.2 (about 7 px, 4 mm at the near limit for $f_x \approx 688$ px), so posts of 2 mm or less. The post-only control sites confirm the post is not detected.

As-built record. Measure each feature's front-face diameter, land width, bevel angle, and position relative to the dowel datum with an optical comparator or calibrated microscope, and measure the reflectance of each surface at the projector wavelength. Record each value with its measurement uncertainty in targets_asbuilt.csv. All analyses use these as-built values, never the nominal ones.